



Master Thesis

**Quantifying Enhanced Construct Specification
using LLMs**

by

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Quantifying Enhanced Construct Specification using LLMs

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

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1 Introduction

What determines the quality of a text? And how do we measure it at scale? These two questions have become increasingly important with the development of advanced Natural Language Generation (NLG) systems such as LLMs that can readily generate any type of text. Although Generative AI (GenAI) promises great potential for industrial applications, its implementation in business contexts is hindered by the inherent variability of output quality.

One company facing this problem is Barter. Barter is an influencer marketing platform, where companies can post collaboration opportunities ("Deals") which content creators can apply to. GenAI is increasingly being used for marketing copy generation, but without adequate quality control, the accumulation of low-quality output becomes a significant operational problem at high volumes.

The need for improvements in the validation of NLG systems is increasingly being recognized [1, 2, 5, 6, 12, 13, 22, 28, 37, 44, 46], calling on the methods from psychometrics to formalize the measurement procedure. Psychometrics concerns itself with the measurement of latent, unobservable variables; indeed, in the social sciences, the variable of interest is rarely directly observable. Meaningful measurement therefore requires specifying how the unobservable target construct relates to the indicators or criteria through which it is measured.

Analogously, in text evaluation, complex constructs such as *Text Quality* or *Deal Attractiveness* are not directly readable properties of a text, but can be decomposed into theoretically related subdimensions that enable fine-grained diagnostics. In a seminal paper, Fabbri et al. [11] decompose "Summary Quality" into the dimensions *coherence*, *consistency*, *fluency* and *relevance*, which they use to show that summary models have specific failure modes: pre-trained language models correlate highly with human judgments on consistency and fluency, whereas extractive models struggled with coherence and relevance. Multi-dimensional evaluation has become the dominant paradigm in automated evaluation [53], and with the introduction of LLMs, has become increasingly flexible.

However, many modern evaluation methods do not assume a specific measurement model. As a result, it remains unclear whether the selected dimensions adequately define the construct, how they should be recomposed into an overall score, and what claims that score supports. This highlights the central problem of this thesis: For artefact-level evaluation, when does measurement become meaningful, and under which conditions?

We argue that artefact-level evaluation can best be understood as a formative measurement problem. In a formative measurement model, the target construct is defined by its constituent dimensions. Because of this, omitting or underspecifying a relevant dimension means omitting or changing part of the construct itself [7]. This creates a central validity risk: holistic scores, or multi-dimensional scores that are weakly specified, might be ill-aligned with the evaluation target.

To test this argument, the thesis compares increasingly specified representations of artefact-level quality: holistic scoring, weakly specified or macro-level decomposition, and fine-grained formative decomposition. In the formative condition, theoretically specified dimensions are measured separately and recomposed explicitly into an overall score. This design addresses two research questions:

- RQ1:** Does stronger construct specification provide stronger validity evidence than holistic or weakly specified alternatives?
- RQ2:** Does stronger construct specification improve the diagnostic interpretability of artefact-level measurements relative to holistic or weakly specified alternatives?

The analysis is conducted in two validation settings. First, the FeedbackQA dataset is used to assess convergent validity: whether LLM-derived measurements of Q&A answer quality align with human quality judgments. Second, the Barter Deals dataset is used to assess predictive validity: whether LLM-derived measurements of deal attractiveness predict aggregate creator application behavior. Before comparing validity evidence, the thesis also diagnoses the LLM measurements themselves, examining scale use, rating uncertainty, verbosity sensitivity, and construct separability. Together, these analyses evaluate whether formative decomposition improves artefact-level LLM measurement not only by producing better predictions, but by yielding scores whose interpretation is more explicitly tied to the construct being measured.

2 Related Work

2.1 From Similarity Metrics to Multi-dimensional Evaluation

Early methods for automated text evaluation centered on lexical overlap, most prominently the BiLingual Evaluation Understudy (BLEU) [33] and Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [27]. Both metrics operationalize quality through surface-level similarity to reference texts: BLEU measures modified n -gram precision against human translations, while ROUGE measures how much reference content is recovered by a candidate summary.

However, the efficacy of lexical-overlap metrics is limited. Surface-level textual features do not capture the rich semantic information inherent to language, limiting their validity as measures of text quality [35]. This motivated embedding-based evaluation methods such as BERTScore [49], BLEURT [38], and MoverScore [50], which use contextual embeddings to estimate semantic similarity between candidate and reference texts.

Yet, similarity-based metrics remain limited because holistic similarity scores may not capture all relevant information needed to assess the evaluation target. As Gehrmann et al. [16] note, quality differences between advanced NLG models are not fully reflected in surface-level textual features. This motivated more fine-grained evaluation methods that decompose the evaluation target into multiple criteria. A central example is SummEval, where Fabbri et al. [11] decompose "summary quality" into *coherence*, *consistency*, *fluency*, and *relevance*, allowing them to identify distinct model failure modes.

2.2 Automated Multi-dimensional Evaluators

The discussion sparked by Fabbri et al. [11] led to the development of automated multi-dimensional text evaluators. Early studies develop dimension-specific evaluators, such as for *factual consistency* in summarization [24, 42], or *coherence* in dialogue generation [10, 48]. Later studies worked towards unified measurement, such as Zhong et al. [53], who propose a method to quantify any arbitrary dimension using transformer models.

Although their method correlates highly with expert annotations, it requires a costly pre-training stage and reference documents. The introduction of LLMs, known for their general problem-solving capabilities, offered a possible solution to these two problems. Early research into using LLMs as reference-free NLG evaluators [15, 43] exhibited lower correlations with expert judgments compared to neural evaluators such as Zhong et al. [53]. However, with the release of GPT-4, Liu et al. [29] outperformed both reference-based and reference-free baselines by a substantial amount. Zheng et al. [52] introduce novel benchmarks to analyse NLG model performance on open-ended NLG tasks and show that LLM evaluations obtain an agreement with human preferences comparable to human-to-human agreement, solidifying the use of LLMs as general-purpose evaluators.

2.3 Textual Quantification in the Social Sciences

The potential of LLMs for quantifying textual attributes is not limited to benchmarking NLG models. Given the function of language as a means for expression, linguistic analysis offers a window into human behavior. Text data analysis is used in psychology to study human behavioral traits, such as personality [8], psychological well-being [19], and emotional states [21]; in economics, to study stock market dynamics using text from news and social media [30]; and in political science, to study partisanship using congressional speeches [17]. Given the abundance of text data and the costliness of manual annotation, NLP methods are

therefore a natural choice to facilitate the measurement of complex constructs from text.

The social sciences faced similar limitations as the NLG community. Dictionary methods, which consider basic features such as word count, are cheap to implement, but often too superficial to answer complex research questions. Representation-based methods, such as embedding models, are expensive to deploy, and most suitable for answering narrow research questions. LLMs again offered the potential for a one-size-fits-all solution: Rathje et al. [34] show that GPT models can accurately measure numerous psychological constructs from texts in multiple different languages, and Le Mens et al. [25] use GPT-4 to quantify semantical properties of texts, both studies achieving state-of-the-art performance. More generally, Yang et al. [47] propose a method to measure arbitrary concepts from text data leveraging the LLMs internal concept representation.

2.4 Conditions for Meaningful Measurement

The current LLM-based methods for measuring constructs from text in the social sciences show many similarities to the evaluation of the capabilities of NLG models in ML: Both fields define measurement targets which they operationalize through LLM scoring. However, as a consequence of their measurement assumptions, they differ in the interpretations that the measurements permit.

The considerations in Fabbri et al. [11] are mostly pragmatic. They observed that a single holistic similarity score did not sufficiently cover the quality differences between summarization systems, so they proposed to measure subdimensions that expose these differences. For evaluating models on these specific criteria, this is sufficient. However, their procedure falls short of permitting general claims about summary quality and summarization capabilities.

As Salaudeen et al. [37] argue, this inferential leap requires justifying the relationship between the measurements and the object of the claim; in this case, the relationship between criterion scores and the construct of "summarization ability". This practice is standard in the social sciences: Boyd and Penebaker [8], Gentzkow et al. [17], Guntuku et al. [19], Kahn et al. [21], Le Mens et al. [25], Manela and Moreira [30], Rathje et al. [34], Yang et al. [47] all operate within a framework where the target constructs are theoretically defined, and where measurements are validated against established measures or theoretically relevant external criteria. Because of this, their measurements are not just criterion-level scores, but construct measurements whose interpretation is anchored in the broader theoretical frameworks of their respective fields.

2.5 Meaningful Multi-Dimensional Measurement

Criticisms of Multi-Dimensional LLM Evaluators To achieve the level of meaningful measurement seen in the social sciences, a growing number of studies are recognizing the need for validity-guided workflows for LLM evaluation based on measurement theory and psychometrics [1, 2, 5, 6, 12, 13, 22, 28, 37, 44, 46].

Yet, advanced benchmarks do not immediately solve the problem of automated evaluation, particularly for open-ended language generation tasks, for which the set of possible correct answers is large and often unknown [14].

As it stands, the focus in automated NLG evaluation still predominantly revolves around theory-agnostic multi-dimensional measurement protocols, where criteria are measured in isolation and the relationship between the measurement and the evaluation target is underspecified. As Arias et al. [3] point out, this holds for 88% of the papers published in NLP evaluation as of 2024, while only 7.6% employ true multicriteria methods, even though linguistic quality is inherently multi-dimensional. This distinction marks the central problem for meaningful multi-dimensional measurement: decomposition is necessary, but not sufficient; additionally, the criteria must be related back to the target construct.

Relating Criteria to Constructs Importantly, this relationship defines the level at which the claims are made: at the system level, or at the artefact level. Benchmarks concern system-level claims, as they attempt to measure some latent model ability. Psychometric methods lend themselves well to this, as many have been developed for this specific purpose. For example, in the same way that latent writing ability can be inferred from grading an essay on multiple criteria, LLMs can be used as scorers to achieve the same goal [45]. However, these methods are not designed for artefact-level claims. Scores that are valid for comparing systems do not automatically support diagnostic claims about individual artefacts. For example, when a model obtains a high score on writing ability, it does not automatically imply that every single text it produces is of high quality, nor does it highlight what needs to be improved about the text. For diagnostic purposes, therefore, different methods are required.

The field has proposed several approaches, with varying measurement assumptions. For automated evaluation of a dialogue generation task, Hashemi et al. [20] propose a method to model rater idiosyncrasies using multi-dimensional LLM scores. They collect data from 24 humans who annotate an average of 31 dialogues on their overall quality. By using criterion-level LLM scores and fitting them to the holistic human ratings using a calibration network, they can accurately model individual rater behavior, enabling them to model for example expert annotators. However, they explicitly take the stance that they model subjective rating behavior, not some ground truth. Kola et al. [23] take a similarly pragmatic approach, where they reuse the same six evaluation criteria for evaluation across multiple distinct evaluation tasks, achieving state-of-the-art alignment with human ratings. They argue that the six-dimensional LLM ratings contain all required information, and a lightweight calibration head is sufficient to extract it.

Meaningful Artefact-Level Diagnostics Hashemi et al. [20] and Kola et al. [23] are examples of the frontier of multi-dimensional evaluation for diagnostic purposes, and they confirm that decomposition has potential for improving LLM evaluators. However, the pragmatism of their methods limits their utility. First of all, they both use calibration heads as their recomposition function. Although this

might improve alignment with human ratings, it lacks explainability. Because of this, although the accuracy of their single predicted holistic quality score is high, it is difficult to determine which dimensions contributed to this score, and how. Moreover, their approaches explicitly model human idiosyncrasies. This is not necessarily wrong; human annotations are often considered the gold standard in alignment research. However, it is also widely acknowledged that human labels are inherently noisy [51], with recent evidence suggesting that LLMs can sometimes outperform expert coders on difficult annotation tasks [41]. Therefore, methods are needed that distinguish alignment with human ratings from meaningful measurement of the target construct, treating human judgments as one source of validity evidence rather than as the object of measurement itself.

Bang et al. [4] take an important step in this direction with UniScore, which derives transparent criterion weights from their discriminative power against an external signal. However, while this makes recombination interpretable, it leaves the measurement model implicit. Because the criteria remain user-defined, the framework can show how strongly each criterion relates to an external signal, but not whether the selected criteria adequately define the construct that the final score is interpreted as measuring. This is a central validity risk for artefact-level diagnostics: when an overall score is constructed from multiple criteria, the choice of criteria is itself part of the construct specification.

The present thesis addresses this problem by treating artefact-level LLM evaluation as a construct-specification problem rather than as just an aggregation problem. It examines whether theoretically grounded criteria, measured separately through LLM prompts and recomposed into an overall score, provide stronger validity evidence and more interpretable artefact-level measurements than holistic or weakly specified alternatives. To formalize this approach, artefact-level scoring is specified as a formative measurement model, drawing on established model specifications from psychometrics.

3 Measurement Framework

This section specifies the validity-guided measurement framework used to evaluate whether LLM-based scores can be interpreted as meaningful measurements. In psychometrics, validity concerns whether theoretical rationales and empirical evidence support the intended interpretation and use of a measurement [32]. Accordingly, the framework has two components: first, a theoretical account of the measurement model assumed for artefact-level diagnostics; and second, an empirical account of how validity evidence for alternative construct specifications can be evaluated.

Section 3.1 defines the measurement model. Drawing on the distinction between reflective and formative specifications [7], we argue that artefact-level diagnostic evaluation is best understood formatively: the target construct is composed of theoretically relevant dimensions rather than inferred from interchangeable indicators. This makes construct underrepresentation and construct underspecification central validity risks. Section 3.2 then defines the role of em-

pirical validation. Because the target constructs are not directly observable, validity cannot be established by comparison to a known true score. Instead, the analyses assess whether increasingly specified construct representations produce stronger relationships with relevant external criteria: human evaluations in the convergent-validity setting and theoretically related behavioral outcomes in the predictive-validity setting.

3.1 Construct Specification and Measurement Models

LLM-based evaluators become measurement instruments when their scores are interpreted as evidence about a target construct, such as answer quality or deal attractiveness. These constructs are not directly observed in the text, but are specified and operationalized through the measurement procedure used to evaluate it. Such an interpretation necessarily implies a measurement model: the criteria must be assumed to relate to the target construct in a particular way. Latent variable models make this assumption explicit by distinguishing whether observed scores are treated as effects of an underlying construct or as constitutive dimensions of a composite construct. The measurement model therefore determines how the criteria are assumed to relate to the construct, and what kinds of validity risks follow from that assumption.

Latent Variable Models Latent variable models define the relationship between a target construct and its indicators. Two model specifications are commonly distinguished: formative and reflective [7]. They differ in their assumptions about the direction of the causal relationship between the indicators ¹ and the construct.

In formative models, the latent variable is assumed to be a composite construct, caused by the indicators. The canonical example is Socioeconomic Status (SES), which can be defined as the composite of three main features: income, education, and occupational prestige [31]. Formally, for a set of causal indicators indexed by $i = 1, 2, \dots, n$, we assume

$$\eta = \sum_{i=1}^n \gamma_i x_i + \zeta \quad (1)$$

where η is a latent construct, γ_i a formative weight, x_i a causal indicator, ζ the disturbance term.

In reflective models, the latent variable causes the indicators. For example, following a "g-factor" interpretation of the latent variable intelligence [39], we assume that intelligence is an inherent trait that humans possess which causes people's performance on cognitive tasks. Formally, for any effect indicator i , we assume

¹ We use the term *indicator* here, following standard psychometric vocabulary. In the context of LLM evaluations, this corresponds to the theoretically defined subdimensions, operationalized as LLM-derived criterion scores.

$$x_i = \lambda_i \eta + \varepsilon_i \quad (2)$$

where x_i is an effect indicator, λ_i a factor loading, η the latent construct, and ε_i the measurement error.

Model Assumptions & Validity Risks In a formative model, the indicators are not assumed to covary because of a shared latent cause. Returning to the example of SES, highly educated people may have low income, and vice versa, even though education, income, and occupational prestige often covary in practice. This has an important consequence: because each indicator uniquely contributes to the value of the latent variable, omitting an indicator means omitting part of the construct [7]. The main validity risks therefore relate to construct underrepresentation: whether the target construct is adequately covered by the indicators defined.

In contrast, in a reflective model, the indicators are treated as manifestations of a shared latent cause. For example, under a "g-factor" interpretation of intelligence, performance across cognitive tasks is assumed to reflect a common underlying trait. Individual items may contain item-specific error, but with a sufficiently broad set of valid indicators, this error can be averaged out when estimating the latent trait. This illustrates the primary consequence of the reflective assumption: the indicators are theoretically interchangeable. The main validity risks therefore concern whether the indicators can plausibly be treated as manifestations of the same latent construct. If they instead reflect multiple constructs, item-specific effects, or construct-irrelevant sources of variation, the reflective measurement model is misspecified.

Artefact-Level and System-Level Measurement Models As discussed in Section 2.5, system-level evaluation concerns the measurement of model abilities through benchmarks. This resembles a reflective specification: the latent trait is the model ability, while benchmark items, tasks, or criteria function as indicators from which that ability is inferred. Such a model is appropriate when the intended claim concerns average performance across a domain of tasks. However, it does not directly support diagnostic claims about individual artefacts. A model may have high latent writing ability, or score well on a writing benchmark, without every text it produces being high quality.

Artefact-level evaluation therefore requires a different measurement specification. When the target is the quality of a specific answer or the attractiveness of a specific deal, the relevant criteria are not interchangeable indicators of a hidden ability. Rather, they define the dimensions along which the artefact is evaluated. Artefact-level diagnostic evaluation is therefore best understood formatively: the overall construct is composed of theoretically relevant dimensions, and omitting or underspecifying these dimensions risks omitting or misspecifying part of the construct itself.

This distinction motivates the experimental setup, defined in Section 4.2 and Section 4.3. If artefact-level quality is formative, then holistic scores are vulnerable to construct underspecification: they produce an overall judgment without

specifying which dimensions constitute that judgment or how they are weighted. Weakly decomposed scores may reduce this problem only partially if the dimensions remain too broad or are not measured independently. The formative condition therefore operationalizes the target construct as a set of theoretically specified dimensions, each measured separately and recomposed explicitly. The next subsection defines how the validity of these alternative construct representations is evaluated empirically.

3.2 Construct Validity and Empirical Validation

Because the target constructs are not directly observable, the formative specification cannot be validated by comparing the resulting scores to a known true value. Instead, this thesis adopts a construct-validity approach, where validity concerns whether evidence supports the intended interpretation and use of a measurement [32]. Human ratings and behavioral outcomes are therefore treated as sources of validity evidence, not as the object of measurement itself. They do not exhaust the meaning of answer quality or deal attractiveness, but provide theoretically relevant external criteria against which the expected behavior of the measurements can be assessed.

The role of empirical validation is therefore to evaluate the consequences of construct representation. The previous subsection identified construct underrepresentation and construct underspecification as central validity risks for formative artefact-level measurement. If these risks matter empirically, then more explicitly specified representations of the target construct should show stronger relationships with relevant external criteria than holistic or weakly specified alternatives. Stronger validity evidence is obtained when theoretically specified criterion measurements explain or predict the external criterion more clearly, more strongly, or more consistently than less specified scores.

Let $\mathbf{x}(d)$ denote the vector of LLM-derived criterion measurements for document d , and let $x^{\text{hol}}(d)$ denote a holistic LLM score for the same document. For a validation criterion $y^{(r)}(d)$, the comparison can be written generically as

$$y^{(r)}(d) \approx f_r(\mathbf{x}(d)) \quad \text{vs.} \quad y^{(r)}(d) \approx h_r(x^{\text{hol}}(d)). \quad (3)$$

The empirical question is not whether either side recovers a true score, but whether the more explicit construct representation produces stronger validity evidence for the intended interpretation of the measurement.

This logic is evaluated in two complementary validation settings. Convergent validity is assessed on FeedbackQA, where human quality judgments provide an external reference for Q&A answer quality. Predictive validity is assessed on Barter Deals, where LLM-derived measurements of deal attractiveness are related to theoretically relevant behavioral outcomes, measured as aggregate creator application behavior. The empirical analyses therefore compare the validity evidence produced by holistic, weakly specified, and formative representations of the target construct. The datasets, construct definitions, and model specifications are introduced in Sections 4.2 and 4.3.

4 Methods

This section is divided into three parts: the general evaluation procedure, the implementation of convergent validity, and the implementation of predictive validity.

The first part describes how the LLM evaluations are obtained and diagnosed. It covers model selection, prompt construction, experimental conditions, logit-based score extraction, and the diagnostic metrics used to assess whether the LLM outputs exhibit favorable properties for formative measurement (Section 4.1.4).

The second part assesses convergent validity using the academic dataset *FeedbackQA* [26]. *FeedbackQA* contains Q&A pairs with human evaluations of answer quality. We define *Q&A Answer Quality* as a formative construct and decompose it into four dimensions. To test convergent validity, we estimate regression models in which the human ratings are regressed on the LLM measurements.

The third part assesses predictive validity using the proprietary industry dataset *Barter Deals*. *Barter Deals* contains written collaboration opportunities for creators, referred to as Deals. In a typical Deal, a company offers a product, service or experience in exchange for promotional creator content. On the Barter marketplace, creators view these offers and apply to the Deals they find appealing. This provides the external behavioral criterion used to assess predictive validity: the total number of applications a Deal receives within its first 7 days of going live (*Apps After 7 Days*). We define *Deal Attractiveness* as a formative construct and decompose it at two levels: first into three macro-dimensions, and then into ten fine-grained dimensions. To test predictive validity, we estimate regression models in which *Apps After 7 Days* is regressed on the LLM measurements and additional market control variables.

4.1 General Evaluation Procedure & Diagnostics

4.1.1 Model selection To perform evaluations we select four popular small-scale, open-source instruction-tuned models: Llama-3.2-3B-Instruct ("Llama 3.2") (Meta), Qwen3-4B-Instruct-2507 ("Qwen 3") [40] (Alibaba), Qwen/Qwen3.5-4B ("Qwen 3.5") (Alibaba), and google/gemma-4-E2B-it ("Gemma 4") (Google). These models were selected to represent recent small-scale open-source instruction-tuned evaluators that are feasible for logit-based evaluation. Notably, Qwen 3 has previously shown to outperform other similar sized open-source LLMs in evaluation tasks [18], which motivates its inclusion alongside the more recent Qwen 3.5.

The use of open-source models is methodologically required because the proposed scoring procedure depends on access to next-token logits. The focus on small instruction-tuned models reflects the operational setting of the industrial partner, where repeated evaluation requires models that are efficient and inexpensive to run at scale.

4.1.2 Prompt Architecture & Experimental Conditions All evaluations use a standardized prompt structure. For each document d and criterion c_i , the model input is defined as the triplet

$$\mathbf{u}_i(d) = [s; c_i; d].$$

where s denotes the system context and task instruction, c_i denotes the criterion specification, and d denotes the target artefact. The three components are defined as follows:

- s (**System Context & Task**): The baseline instruction that defines the evaluator role, the application setting, and the general evaluation task.
- c_i (**Criterion Specification**): The prompt-level measurement target. This includes the semantic definition of the construct or criterion being evaluated, together with the Likert-scale anchors used to define the rating levels.
- d (**Target Document**): The artefact to be evaluated, such as a Q&A pair in FeedbackQA or the relevant Deal text in Barter Deals.

Across experimental conditions, the system context s and target document d are held constant. The experimental manipulation is implemented through the content and structure of the criterion specification c_i . The experimental design differs per dataset; for *FeedbackQA*, see Section 4.2.2, for *Barter Deals*, see Section 4.3.2.

4.1.3 Logit-Based Score Extraction After constructing the evaluation triplet $\mathbf{u}_i(d)$, we instantiate and supply it to the LLM in Chat Template format, consisting of successive turns with fixed roles. The *system* turn contains s and c_i ², and the *user* turn contains d .

Generation begins at the *assistant* turn, which we manually prefix with the literal string **Rating:** to condition the next-token distribution toward the evaluation tokens (see Appendix A for exact templates). We then perform a single forward pass to obtain the next-token logits $\mathbf{z}(\mathbf{u}_i) \in \mathbb{R}^{|\mathcal{V}|}$.

We restrict the logits to the subset of rating tokens $\mathcal{S} = '1', \dots, 'K'$, where K represents the maximum integer of the specified Likert scale. This yields the restricted logit vector $\mathbf{z}_{\mathcal{S}}(\mathbf{u}_i) = z_k(\mathbf{u}_i) * k \in \mathcal{S}$.

The probability distribution over the rating tokens is obtained by applying a softmax to the restricted logit vector:

$$P_{\mathcal{S}}(k | \mathbf{u}_i) = \frac{\exp(z_k(\mathbf{u}_i))}{\sum_{*j \in \mathcal{S}} \exp(z_j(\mathbf{u}_i))}. \quad (4)$$

This defines a discrete probability distribution over $\mathcal{S}$, from which we derive two statistics: the expected value (EV), used as the evaluation score, and the normalized Shannon entropy, used as a diagnostic of distributional spread.

² For the informed baseline condition, c_i represents a composite set of criteria rather than an isolated dimension; see Section 4.2.2.

The expected value represents the LLM’s evaluation for a given prompt-document pair, where lower values correspond to lower ratings on the given criterion and higher values correspond to higher ratings. The EV is used as the primary LLM-derived measurement in the validation models and is computed as

$$x_i^{\text{EV}} = \sum_{k \in \mathcal{S}} v(k) \cdot P_{\mathcal{S}}(k \mid \mathbf{u}_i), \quad (5)$$

where x_i^{EV} is the EV score, and $v(k)$ is a function that maps token string k to its corresponding integer value. This yields a continuous score that uses the probability mass assigned to each rating category. EV ratings are z-standardized within each model, dataset, condition, and criterion before entering the regression models to facilitate inter-model comparisons.

The normalized Shannon entropy captures how probability mass is distributed across the rating tokens. Low entropy indicates that probability mass is concentrated on one or a few tokens, while high entropy indicates that probability mass is distributed more evenly across the rating scale. Entropy is used for diagnostic purposes, as described further in Section 4.1.4.

To make entropy comparable across scales with different numbers of categories, such as the four-point scale used for FeedbackQA and the seven-point scale used for Barter Deals, we apply a normalization factor. Let $K = |\mathcal{S}|$ denote the number of valid rating categories. The normalized entropy H_i for prompt sequence $\mathbf{u} * i$ is defined as:

$$H_i = -\frac{1}{\ln(K)} \sum_{k \in \mathcal{S}} P_{\mathcal{S}}(k \mid \mathbf{u} * i) \ln P_{\mathcal{S}}(k \mid \mathbf{u}_i). \quad (6)$$

This specification takes values in $[0, 1]$, attaining 0 when all probability mass is concentrated on a single token and 1 when the distribution is uniform.

4.1.4 Instrument Diagnostics Below, we outline the diagnostics used to characterize the LLM-derived measurements. Additional details are provided in Appendix D.

Distributional Properties of Evaluations and Entropy To empirically diagnose the sensitivity of the LLMs as continuous measurement instruments, we analyze the first two moments (mean and standard deviation (SD)) of both the EV ratings and the normalized Shannon entropy (Table 3). These summary statistics provide a preliminary diagnostic of whether the logit-derived metrics exhibit the properties expected from a continuous measurement instrument: The EV dispersion indicates scale utilization, and the entropy indicates whether the distribution that produced the EV scores is concentrated around single integers or evenly distributed across the rating scale. More detailed interpretations on rating regimes are provided in Appendix D.1.

Diagnosing Residual Verbosity Sensitivity LLMs have been shown to prefer longer, more verbose documents over shorter documents even when quality is comparable, a phenomenon known as verbosity bias [36]. This is problematic for measurement, since a valid instrument should respond to the specified criterion rather than to construct-irrelevant variation in text length. We therefore diagnose whether LLM ratings remain sensitive to text length after documents have reached a mature length. We refer to this broader diagnostic as *residual verbosity sensitivity*.

To quantify residual verbosity sensitivity, we compute the absolute Cohen’s d difference in mean EV ratings between documents in Q3 and Q4 of the text-length distribution. The diagnostic is computed separately by model and criterion. Larger values indicate stronger residual sensitivity to length among relatively long documents. Following Cohen [9], we interpret $|\delta| \approx 0.2$ as small, $|\delta| \approx 0.5$ as medium, and $|\delta| \geq 0.8$ as large. For specifics, see Appendix D.2.

Construct Separability and Multicollinearity We diagnose construct separability using VIF, interpreting values above 5 as moderate and values above 10 as substantial multicollinearity. For specifics, see Appendix D.3.

4.2 Convergent Validity: FeedbackQA

4.2.1 Dataset Overview The *FeedbackQA* dataset Li et al. [26] contains evaluations of Q&A pairs ($n = 9065$) related to COVID-19, which were scraped from the websites of WHO, US Government, UK Government, Canadian Government, and the Australian Government. Crowdsourced workers from English-speaking countries were hired through Amazon’s MTurk platform to rate each Q&A pair and provide feedback on their rating. The rating is performed on a four-point Likert scale (*bad*, *could be improved*, *good*, *excellent*). The feedback concerns "a natural language explanation elaborating on the strengths and/or weaknesses of the answer" [26], which we leverage to determine the formative indicators.

The text length of the Q&A pairs has a mean of 169.09 words with a standard deviation of 148.93, with the full empirical distribution visualized in Appendix C.1. The mature text brackets used for the verbosity-bias diagnostic are defined from the corpus-specific length distribution, where the median word count is 126 words and the 75th percentile is 202 words.

The distribution of the annotation-level human ratings is shown in Table 1. The distribution is heavily bimodal, with the majority of the ratings either 1 or 4 (65.21%), while the intermediate ratings of 2 or 3 account for the remainder (34.79%).

4.2.2 Definition of Experimental Conditions and Evaluation Criteria

We define three experimental conditions: the naive baseline, the informed baseline, and the formative experimental condition. The naive baseline evaluates the holistic construct of *Overall Answer Quality* using a minimal semantic definition.

The formative condition is constructed by deriving dimensions empirically from the natural language explanations provided by the crowdsourced raters,

Table 1: Distribution of Human Ratings for FeedbackQA

Rating	Count (Annotations)	Percentage (%)
1 - Bad	6086	33.57
2 - Could be improved	3120	17.21
3 - Good	3188	17.58
4 - Excellent	5736	31.64

supported by the thematic analysis conducted by Li et al. [26]. These explanations indicate that reviewers primarily evaluate whether answers are relevant, complete, focused, and clearly expressed. We therefore decompose *Q&A Answer Quality* into four formative dimensions: *Topical Alignment*, *Information Coverage*, *Topical Concentration*, and *Communicative Clarity*. The full criterion definitions are provided in Appendix E.

The informed baseline uses the same enriched descriptions of the four dimensions, but requires the LLM to evaluate them jointly and output a single holistic score. This intermediate baseline distinguishes the effect of enriched construct specification from the additional effect of independently measuring each formative dimension.

4.2.3 Model Specification

Convergent Validity To assess convergent validity, we regress human-annotated Q&A quality on the LLM-derived measurements, treating the four-point Likert ratings as continuous interval data. Because each document d has one LLM evaluation but two human ratings indexed by $j \in \{1, 2\}$, the regression is estimated at the annotation level. Each document-level LLM measurement is therefore paired separately with each human rating y_{dj}^H . To control for verbosity bias, all specifications include the natural logarithm of the document word count, $\ln(W_d)$.

For the formative condition, we estimate:

$$y_{dj}^H = \beta_0 + \sum_{k \in K_F} \beta_k x_{k,d} + \delta \ln(W_d) + \varepsilon_{dj}, \quad (7)$$

where y_{dj}^H denotes the human quality rating assigned to document d in annotation j . The set K_F contains the four formative dimensions. The predictor $x_{k,d}$ denotes the continuous LLM EV measurement for criterion k and document d .

For the naive and informed baselines, the model is restricted to a single holistic LLM measurement:

$$y_{dj}^H = \beta_0 + \beta_1 x_{\text{holistic},d} + \delta \ln(W_d) + \varepsilon_{dj}. \quad (8)$$

Here, $x_{\text{holistic},d}$ denotes the condition-specific holistic EV measurement for document d : the minimally specified *Overall Answer Quality* score in the naive baseline and the enriched holistic answer-quality score in the informed baseline.

Because the LLM measurements and word counts vary at the document level, while the human ratings vary at the annotation level, residuals may be correlated within documents. We therefore estimate all models using Ordinary Least Squares (OLS) with standard errors clustered at the document level.

Cross-Validated Alignment Metrics For each specification, we generate out-of-fold predictions and compute RMSE, Pearson’s r , Spearman’s ρ , and Kendall’s τ between predicted and observed ratings. Five-fold cross-validation is grouped by document, ensuring that the two annotations for the same document never appear in both the training and test sets. Human-human alignment is reported as a reference benchmark for annotation consistency, not as a competing model.

4.3 Predictive Validity: Barter Deals

4.3.1 Dataset Overview The *Barter Deals* dataset contains *Deals* written by companies ($n = 3137$). Barter is an online marketplace where companies (*partners*) can barter goods and services in exchange for influencer content. For example, a company that sells hair styling tools offers their new product in exchange for a video of a creator using it; or, a museum provides free entrance to a creator and their friends in exchange for a video of their visit. Creators view Deals on the Barter app, and can apply to the deals for which they are eligible.

The *Deal Text* is the marketing copy that is posted on the Barter platform as part of the Deals that creators can apply to, and consists of a Title, Body, and Requirements. The Title provides a quick overview of the deal. The Body contains persuasive marketing copy used to inform the creator about details of the product, service, or experience being offered, as well as optional additional rewards. The Requirements section specifies what the company requires from the creator, such as the number of videos, script constraints, or details on the intended advertising use of the content.

The outcome variable is the number of applications a Deal receives within the first seven days of going live (*applications after 7 days*). This variable is used as a behavioral criterion for *Deal Attractiveness*. Although application volume is determined by both textual quality and market structure, meaningful differences in the attractiveness of Deals are expected to be partially reflected in creator application behavior.

Corpus-specific text-length statistics and quartile thresholds for the Barter verbosity diagnostic are reported in Appendix C.1.

Table 2: Descriptive statistics of the behavioral KPI *applications after 7 days*

Variable	N	Mean	SD	Min	P25	Median	P75	Max
Applications after 7 days	3137	9.48	16.97	0	0	3	11	188

The application count is highly dispersed with substantial mass at zero (25th percentile = 0), motivating a count model that accounts for overdispersion, such as a Negative Binomial.

4.3.2 Definition of Experimental Conditions and Evaluation Criteria

We define three experimental conditions: the naive baseline, the macro-formative baseline, and the formative condition. All dimensions across these conditions are measured on a seven-point Likert scale to allow finer-grained variation in the evaluations.

While the informed baseline in Section 4.2 tests the general efficacy of structural decomposition, the macro-formative baseline used for Barter Deals isolates the value of fine-grained decomposition relative to broader macro-level evaluations commonly used in NLP [11, 29?].

The naive baseline evaluates the holistic construct of *Overall Deal Attractiveness*, measuring "the overall appeal of the barter opportunity from the perspective of a typical content creator."

The macro-formative baseline serves as intermediate condition, and jointly measures the three theoretical pillars of *Deal Attractiveness* established in Section ??: Reward, Cost, and Pitch Quality.

The formative condition decomposes the three macro-dimensions into ten fine-grained dimensions: *Brand Prestige*, *Perceived Economic Value*, *Creator Brand Alignment*, *Functional Utility*, *Product Universality*, *Workload Volume*, *Creative Restrictiveness*, *Call-to-Action Strength*, *Rhetorical Objectivity*, and *Proposition Clarity*. The full criterion definitions are provided in Appendix E.

4.3.3 Model Specification

Predictive Validity To assess predictive validity in a real-world behavioral setting, we model creator application volume using the Barter Deals dataset. The dependent variable, *applications after 7 days*, is an overdispersed count outcome, so we estimate Negative Binomial regression models with a log link. Because multiple Deals may be posted by the same brand, all models use standard errors clustered at the partner ID level.

Before introducing the LLM-derived measurements, we estimate a structured base model containing only non-LLM platform variables:

$$\ln(\mu_i) = \beta_0 + \gamma \mathbf{Z}_i + \alpha_t, \quad (9)$$

where μ_i is the expected application count for Deal i . The vector $\mathbf{Z}_i$ contains structural controls for strict follower requirements (defined by Deals that require creators to have > 4000 followers in order to be eligible to apply), competitor density, platform strategy, geographic market scope, and product category. To account for verbosity-related length effects, $\mathbf{Z}_i$ also includes separate quartile-based indicators for body length and requirements length. This allows length effects to vary across text-length brackets rather than assuming a linear relationship between text length and the outcome variable. Finally, α_t denotes

month-year fixed effects, which absorb platform growth, seasonal variation, and macro-level temporal shocks. The purpose of this base specification is to partial out predictable market structure before evaluating the LLM-derived measurements.

The base model is then augmented with the LLM-derived measurements to test whether they explain additional variation in creator applications. For a given set of LLM predictors K , the augmented specification is:

$$\ln(\mu_i) = \beta_0 + \sum_{k \in K} \beta_k x_{k,i} + \gamma \mathbf{Z}_i + \alpha_t, \quad (10)$$

where $x_{k,i}$ denotes the continuous LLM Expected Value (EV) measurement for criterion k and Deal i .

The set K varies by measurement condition. In the naive baseline, K contains only the holistic *Overall Deal Attractiveness* score. In the macro-formative baseline, K contains the three macro-pillars: Reward, Cost, and Pitch Quality. In the formative condition, K contains the ten fine-grained Deal Quality dimensions.

Model Comparison Criteria Model performance is evaluated using both in-sample likelihood-based criteria and out-of-sample cross-validation. For the in-sample comparison, we report the log-likelihood, McFadden pseudo- R^2 , AIC, and BIC to assess whether the LLM-augmented specifications improve model fit beyond the structured baseline while accounting for model complexity.

To assess generalization, all specifications are also evaluated using 5-fold cross-validation. In each fold, the model is estimated on the training partition and evaluated on held-out Deals. Out-of-sample performance is summarized using RMSE, Poisson deviance, and Spearman’s ρ . RMSE captures prediction error on the original application-count scale, Poisson deviance provides a count-appropriate predictive loss, and Spearman’s ρ captures rank-order alignment between observed and predicted application counts.

For the likelihood-based comparisons, improvements are reported relative to the structural baseline. ΔLL is defined as $LL_{\text{model}} - LL_{\text{baseline}}$, while ΔAIC and ΔBIC are defined as $AIC_{\text{baseline}} - AIC_{\text{model}}$ and $BIC_{\text{baseline}} - BIC_{\text{model}}$, respectively. All three quantities are oriented so that positive values indicate improvement over the structural baseline.

5 Results

The results follow the structure introduced in the Methods chapter. We first report the measurement diagnostics that apply to both datasets: scale utilization and entropy calibration, residual verbosity sensitivity, and criterion separability. These diagnostics are presented jointly for FeedbackQA and Barter Deals because they evaluate the behavior of the LLM-based measurement procedure itself, before assessing its validity.

After establishing these instrument-level properties, we turn to the dataset-specific validation analyses. For FeedbackQA, we assess convergent validity by

comparing LLM-derived evaluations with human quality judgments. For Barter Deals, we assess predictive validity by testing whether the LLM-derived evaluations explain variation in creator application behavior.

5.1 Instrument Diagnostics

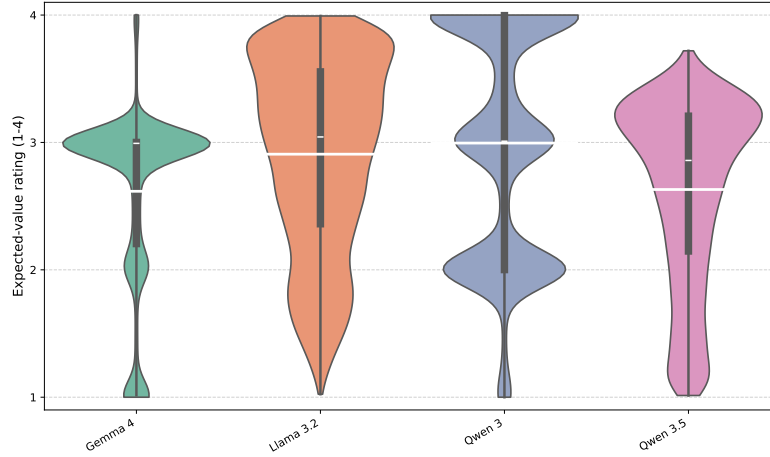


Fig. 1: Expected-value rating distributions for the FeedbackQA *Overall Quality (Naive)* criterion. The figure illustrates representative model-specific rating regimes: Gemma 4 and Qwen 3 show more integer-anchored distributions, whereas Llama 3.2 and Qwen 3.5 produce smoother sub-integer variation. Full criterion-level distributions are reported in Appendix C.

Expected-Value and Entropy Distributions The models exhibit recognizable rating regimes, although these regimes are partly moderated by task and criterion. Figure 1 shows the distribution of EV ratings for the FeedbackQA *Overall Quality (Naive)* criterion as a representative visual diagnostic, while Table 3 summarizes the corresponding dataset-level diagnostics.

Qwen 3.5 exhibits the most favorable profile for continuous measurement. Across both datasets, it combines moderate-to-broad EV dispersion with consistently high entropy, indicating that its ratings vary between integer values rather than collapsing into hard ordinal classifications. Llama 3.2 shows a similar high-entropy regime for FeedbackQA, but in Barter Deals its EV dispersion is substantially lower, indicating a compressed continuous regime.

Qwen 3 and Gemma 4 show lower average entropy and therefore behave more like ordinal classifiers than continuous measurement instruments. Qwen 3 is the most consistent example of this pattern: across both datasets, it combines broad

EV dispersion with low entropy, indicating a broad but quantized rating regime. Gemma 4 is also relatively dispersed, but its entropy is higher in Barter Deals than in FeedbackQA, suggesting a more semi-quantized broad regime in that setting. Notably, the visual distributions further distinguish the two low-entropy models: Qwen 3 forms several distinct integer-centered peaks, whereas Gemma 4 concentrates more mass around preferred rating regions with thinner tails across the scale. Full criterion-level distributions are reported in Appendix C.2.

Table 3: Scale Utilization and Entropy Calibration by Dataset and Model Architecture

Model	Rating: Mean (SD)	Entropy: Mean (SD)
FeedbackQA (1–4 scale)		
Gemma 4	2.54 (0.74)	0.13 (0.18)
Llama 3.2	2.72 (0.77)	0.72 (0.23)
Qwen 3	2.95 (0.88)	0.12 (0.16)
Qwen 3.5	2.66 (0.67)	0.73 (0.15)
Barter Deals (1–7 scale)		
Gemma 4	3.54 (1.00)	0.30 (0.17)
Llama 3.2	4.29 (0.40)	0.67 (0.08)
Qwen 3	4.14 (1.01)	0.18 (0.13)
Qwen 3.5	3.68 (0.79)	0.66 (0.09)

Note. Values are first computed within each evaluation criterion and then averaged across criteria. Ratings are expected-value (EV) scores computed over the model’s rating-token distribution. Rating SD reflects average within-criterion scale utilization: higher values indicate that, for a typical criterion, the model spreads EV scores more broadly across documents. Entropy indicates how concentrated the model’s probability mass is over the rating options: low entropy concentrates EV ratings around integer values, while higher entropy spreads probability across multiple ratings, allowing EV scores to fall between integer values.

Verbosity Sensitivity Table 4 shows that residual verbosity sensitivity is strongly model-dependent. Llama 3.2 is the only model with systematic medium-sized effects in both datasets: all six FeedbackQA comparisons fall in the medium range, and Barter Deals contains eight medium and one large effect. This indicates that Llama 3.2 remains sensitive to text length even after comparing only mature-length texts. By contrast, Gemma 4, Qwen 3, and Qwen 3.5 show mostly negligible or small effects, with no medium or large effects in either dataset. This implies that robustness to text length should be treated as part of model selection, since models with favorable rating regimes may differ in their susceptibility to residual verbosity effects.

Table 4: Summary of Verbosity-Bias Diagnostics by Dataset and Model Architecture

Model	Mean $ d $	Max $ d $	N	Negligible	Small	Medium	Large
FeedbackQA							
Gemma 4	0.16	0.20	6	5	1	0	0
Llama 3.2	0.57	0.62	6	0	0	6	0
Qwen 3	0.15	0.25	6	4	2	0	0
Qwen 3.5	0.11	0.25	6	5	1	0	0
Barter Deals							
Gemma 4	0.17	0.46	14	8	6	0	0
Llama 3.2	0.53	0.85	14	1	4	8	1
Qwen 3	0.20	0.47	14	8	6	0	0
Qwen 3.5	0.17	0.43	14	10	4	0	0

Note. Verbosity bias is measured as absolute Cohen's d from the Q3–Q4 word-count comparison. N is the number of evaluated criterion–text-section comparisons per model. Effect-size bins are mutually exclusive: negligible ($|d| < .20$), small ($.20 \leq |d| < .50$), medium ($.50 \leq |d| < .80$), and large ($|d| \geq .80$). Absolute values are used because length effects may occur in either direction depending on criterion valence.

Construct Separability and Multicollinearity The FeedbackQA VIF diagnostics indicate moderate to substantial multicollinearity among the formative criterion scores. All model specifications contain at least three predictors with VIF values above 5, and Llama 3.2 and Qwen 3.5 also contain predictors above 10. This suggests that the FeedbackQA criteria are empirically difficult to separate, despite being theoretically specified as distinct formative components. Consequently, the FeedbackQA regressions are interpreted primarily in terms of overall model fit and convergent validity, rather than as strong evidence for independent criterion-level effects.

For Barter Deals, the VIF diagnostics indicate limited multicollinearity across all formative specifications. No predictor exceeds the threshold of 5 in any model, and the maximum VIF remains low across architectures, ranging from 1.79 for Gemma 4 to 3.70 for Llama 3.2 in the Formative condition, and from 1.34 for Qwen 3 and Qwen 3.5 to 4.20 for Llama 3.2 in the Macro-Formative condition.

Because the Barter Deals regressions also include non-LLM control variables, we additionally inspected VIFs for the baseline control specification. This diagnostic identified one marginally flagged dummy variable, *macro_category_Home Cooking & CPG*, with a VIF of 5.09. Since this value only slightly exceeds the threshold of 5 and no predictor exceeds 10, the control specification does not indicate substantial multicollinearity.

This confirms that both the formative criteria and the structural covariates provide sufficiently separable information for coefficient-level interpretation in

Table 5: VIF Diagnostics for FeedbackQA Formative Specifications

Group	Max VIF	Mean VIF	N VIF > 5	N VIF > 10	Multicollinearity
Gemma 4 Formative	6.75	4.46	3	0	Moderate
Llama 3.2 Formative	16.93	9.43	4	2	Substantial
Qwen 3 Formative	9.38	6.86	4	0	Moderate
Qwen 3.5 Formative	11.39	6.88	4	1	Substantial

Note. VIF values are computed for the formative criterion-level EV predictors in each model-specific FeedbackQA regression. Values above 5 are interpreted as moderate multicollinearity and values above 10 as substantial multicollinearity. High VIFs indicate that the formative criteria contain overlapping information, limiting independent coefficient-level interpretation.

the predictive validity models. The full Barter Deals VIF diagnostics, including the baseline controls, are reported in Appendix C.3.

5.2 Convergent Validity: FeedbackQA

EV-Based Model Comparison Table 6 reports the main cross-validated convergent-validity results for the EV-only specifications. The full set of cross-validation metrics, including Pearson’s r and Kendall’s τ , is reported in Appendix C.4. Across all model architectures, the formative condition improves over both the naive holistic and informed holistic baselines on RMSE and Spearman’s ρ . The improvements are modest in absolute size, but consistent across models and metrics. The informed holistic condition provides only marginal improvement over the naive holistic condition. This indicates that the benefits of formative decomposition are obtained through isolated criterion measurement, in addition to enhanced construct specification.

The strongest performance is obtained by the Qwen models. Qwen 3 and Qwen 3.5 perform nearly identically in the formative condition, reaching Spearman correlations of 0.578 and 0.576, respectively. Both exceed the human-human baseline on Spearman’s ρ (0.541), although the human-human baseline should be interpreted as a reference point rather than a fitted model. Llama 3.2 performs weakest across conditions, while Gemma 4 shows intermediate performance.

Coefficient-Level Interpretation The formative coefficients are not interpreted substantively for FeedbackQA. As shown in the construct-separability diagnostics, the FeedbackQA criteria exhibit moderate to substantial multicollinearity, meaning that criterion-level coefficients are unstable and do not support clean independent interpretation. This is consistent with the empirical structure of the task: the criteria appear to capture overlapping aspects of general answer quality rather than clearly separable formative components. The FeedbackQA analysis is therefore interpreted at the level of experimental condition performance, fo-

Table 6: Cross-Validated FeedbackQA Performance by Experimental Condition

Metric	Condition	Gemma 4	Llama 3.2	Qwen 3	Qwen 3.5
RMSE ↓	Holistic Naive	1.119	1.144	1.044	1.033
	Holistic Informed	1.111	1.144	1.039	1.034
	Formative	1.094	1.128	1.018	1.019
Spearman's ρ ↑	Holistic Naive	0.436	0.395	0.544	0.559
	Holistic Informed	0.450	0.396	0.550	0.558
	Formative	0.480	0.423	0.578	0.576
Human-Human Baseline		RMSE = 1.196; Spearman's ρ = 0.541			

Note. Values report out-of-fold cross-validated performance for EV-only specifications. RMSE measures prediction error, while Spearman's ρ measures rank alignment with human quality judgments. Lower RMSE and higher Spearman's ρ indicate better convergent validity. Bold values indicate the best-performing model architecture within each metric-condition row. The human-human baseline compares the two available human annotations and is not a fitted model.

cusing on whether formative decomposition improves convergent validity over holistic prompting.

5.3 Predictive Validity: Barter Deals

This section reports the results of the Negative Binomial models predicting the behavioral KPI *applications after 7 days*. The main analysis tests whether the LLM-derived measurements explain variation in creator applications beyond a comprehensive set of observed structural covariates from the Barter database, and whether formative decomposition improves predictive validity relative to holistic measurement. The section therefore focuses on in-sample likelihood-based model comparison and out-of-sample predictive performance. Detailed regression outputs, including coefficient estimates and significance tests, are reported in Appendix C.5.

Structural Baseline Model Before evaluating the LLM-derived measurements, a structural baseline model is estimated using only non-LLM platform covariates. This model serves as the benchmark against which the incremental contribution of the LLM measurements is assessed. Table 7 reports both its in-sample likelihood-based fit and its out-of-sample cross-validated performance.

The baseline model achieves a McFadden pseudo R^2 of 0.056, indicating modest but non-trivial explanatory power given the noisiness of behavioral application data. The estimated dispersion parameter, $\alpha = 0.8590$, is consistent with substantial overdispersion in the application counts, supporting the use of a Negative Binomial specification rather than a Poisson model. As reported in the full regression output in Appendix C.5, the most substantively important baseline effects are associated with market scope, month-year fixed effects,

and product category, consistent with theoretical expectations of marketplace dynamics.

The cross-validated baseline performance shows that exact application counts are difficult to predict using structural platform variables alone, with an RMSE of 19.48. However, the Poisson deviance of 14.68 and Spearman’s ρ of 0.533 indicate that the baseline retains useful count-loss and rank-order information. These values define the out-of-sample reference point for evaluating whether the LLM-derived measurements improve prediction beyond the structural covariates.

Table 7: Structural Baseline Model Performance for Barter Deals

Evaluation	Metric	Value
In-sample fit	Pseudo R^2	0.056
	LogLik	-10489.2
	AIC	21056.4
	BIC	21290.1
Cross-validated performance	RMSE	19.48
	Poisson deviance	14.68
	Spearman’s ρ	0.533

Note. The structural baseline is a Negative Binomial model containing only non-LLM platform covariates, including the controls and fixed effects specified in the methods section. In-sample fit statistics provide the reference point for the likelihood-based LLM comparisons in Table 8; cross-validated performance metrics provide the reference point for the out-of-sample comparisons in Table 9.

Incremental Predictive Validity of LLM Measurements The in-sample results for model fit are reported in Table 8, and the out-of-sample cross validation results are reported in Table 9. Across both in-sample and out-of-sample comparisons, the formative condition provides the strongest and most consistent support for incremental predictive validity. The gains are largest for likelihood-based fit, Poisson deviance, and rank-order performance, while improvements in RMSE are more modest.

In the in-sample comparison, the formative specifications produce substantially larger improvements in log-likelihood, AIC, and BIC than either the Holistic Naive or Macro-Formative specifications. The Holistic Naive and Macro-Formative conditions yield negligible to modest gains, and Llama 3.2 even worsens BIC under the Holistic Naive condition ($\Delta\text{BIC} = -4.0$). By contrast, the formative specifications produce large positive ΔBIC values, ranging from 157.5 for Llama 3.2 to 352.4 for Qwen 3.5. This suggests that fine-grained formative decomposition provides incremental predictive information beyond holistic or macro-level measurements of deal quality.

Table 8: In-Sample Barter Deals Model Comparison by Experimental Condition

Metric	Condition	Gemma 4	Llama 3.2	Qwen 3	Qwen 3.5
Pseudo $R^2 \uparrow$	Holistic Naive	0.058	0.056	0.060	0.059
	Macro-Formative	0.057	0.058	0.062	0.059
	Formative	0.074	0.067	0.072	0.075
$\Delta LL \uparrow$	Holistic Naive	20.4	2.0	43.0	34.5
	Macro-Formative	17.4	23.5	64.8	33.7
	Formative	207.1	118.7	177.1	216.2
$\Delta AIC \uparrow$	Holistic Naive	38.8	2.0	84.1	67.1
	Macro-Formative	28.8	41.0	123.5	61.3
	Formative	394.2	217.5	334.1	412.3
$\Delta BIC \uparrow$	Holistic Naive	32.8	-4.0	78.1	61.1
	Macro-Formative	10.8	23.0	105.5	43.3
	Formative	334.2	157.5	274.2	352.4

Note. Values report in-sample Negative Binomial model comparison for EV-only LLM-augmented specifications. The structural baseline has pseudo $R^2 = 0.056$, $LL = -10489.2$, $AIC = 21056.4$, and $BIC = 21290.1$. ΔLL , ΔAIC , and ΔBIC are computed relative to this structural baseline and are oriented so that positive values indicate improvement. ΔLL is defined as $LL_{\text{model}} - LL_{\text{baseline}}$, while ΔAIC and ΔBIC are defined as $AIC_{\text{baseline}} - AIC_{\text{model}}$ and $BIC_{\text{baseline}} - BIC_{\text{model}}$, respectively. Bold values indicate the best-performing LLM-augmented model family within each metric-condition row.

In the out-of-sample cross-validation results, the smallest improvements are observed in RMSE, which remains high for all conditions, with a value between 18.86 (Qwen 3.5 Formative) and 19.52 (Llama 3.2 Holistic Naive). The clearest out-of-sample improvements are obtained in count-loss and rank-order performance. Although the Holistic Naive and Macro-Formative conditions remain close to the baseline Poisson deviance of 14.68, the Formative condition meaningfully improves it for all models. Gemma 4 obtains the lowest Poisson deviance of 13.23, which is a 9.88% reduction in count-prediction loss relative to the structural baseline. Similarly, the Holistic Naive and Macro-Formative conditions show marginal improvements in rank-order performance compared to the baseline, whereas rank-order performance substantially improved for all models in the Formative condition. Gemma 4 again obtains the best result with $\rho = 0.600$. These results indicate that the LLM-augmented models can meaningfully rank deals by application count, even if they struggles to predict exact application counts.

Comparing the models on the best-performing Formative condition, Gemma 4 and Qwen 3.5 obtain the best overall results, with marginal between-model differences. Llama 3.2 exhibits the smallest improvements, while Qwen 3 obtains intermediate results. The result for Qwen 3.5 is in line with the results for Feed-

Table 9: Cross-Validated Out-of-Sample Predictive Performance by Experimental Condition

Metric	Condition	Gemma 4	Llama 3.2	Qwen 3	Qwen 3.5
Poisson Dev. ↓	Holistic Naive	14.60	14.75	14.46	14.41
	Macro-Formative	14.64	14.46	14.38	14.48
	Formative	13.23	13.91	13.53	13.28
RMSE ↓	Holistic Naive	19.47	19.52	19.41	19.35
	Macro-Formative	19.51	19.30	19.42	19.35
	Formative	18.88	19.08	18.98	18.86
Spearman's ρ ↑	Holistic Naive	0.539	0.530	0.548	0.545
	Macro-Formative	0.536	0.538	0.555	0.539
	Formative	0.600	0.574	0.590	0.594

Note. Values report 5-fold cross-validated out-of-sample predictive performance for EV-only LLM-augmented Negative Binomial specifications. The cross-validated structural baseline achieves Poisson Dev. = 14.68, RMSE = 19.48, and Spearman's ρ = 0.533. Bold values indicate the best-performing LLM-augmented model family within each metric-condition row.

backQA, where it showed the strongest convergent validity. Gemma 4, on the other hand, did not perform well on convergent validity, while it matches the performance of Qwen 3.5 for predictive validity. This indicates that an LLM's measurement performance is not guaranteed to generalize across distinct validity assessments.

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A Prompt rendering via `apply_chat_template`

TODO: CHANGE THIS SO IT MATCHES CURRENT PROMPT

We format each prompt using Hugging Face's chat template with `add_generation_prompt=True`, and then append an explicit assistant prefix (shown below). The rendered template is passed to the model, which computes the next-token probabilities after the assistant prefix. The model input is the concatenation of:

(i) system message, (ii) user message, (iii) assistant header inserted by the template, (iv) `assistant_prefix`.

```
<System role header as produced by tokenizer>
Instruction: Rate the Relevance of the Answer provided for
the Question
on a scale of 1 to 4. Use the following rubric to determine
the score:

Relevance: Does the answer discuss the specific topic and
entities requested?

1 = Irrelevant: Discusses a completely different topic (e.g.,
"baby delivery"
instead of "respirators").

2 = Topic Mismatch: Discusses a related category but the
wrong specific
entity (e.g., "Work Visa" instead of "Pandemic Visa").

3 = Broadly Relevant: Discusses the correct topic but
includes significant
tangential or unrelated information.

4 = Precise: Focuses exclusively on the specific entity and
topic requested
in the question.
</System role header>

<User role header as produced by tokenizer>
Question:
{question}

Answer:
{answer}
</User role header>

<Assistant role header inserted by add_generation_prompt=True
>
Rating:
```


B Sample data

B.1 FeedbackQA examples

Example F01

Question: Is it still possible to access medical appointments and medications through my doctor?

Answer: You can access bulk-billed appointments with your doctor, nurse or mental health professional via phone or videoconference. If you would like to arrange a telehealth appointment, contact your doctor. Accessing medicines while staying at home. Your doctor may be able to arrange to have your prescriptions sent to your pharmacy directly so you can stay at home. For more information on whether you can access this service please call your pharmacy and speak to them about the best way to arrange this.

Rating 1: 4

Feedback 1: Gives information on how patients can access appointments with their doctors during the pandemic.

Rating 2: 3

Feedback 2: This answer relates to accessing medical appointments and medications, but does not give much information.

Table 10: Sampled FeedbackQA items (IDs and labels), and Qwen scores for Relevance, Completeness, Directness, and Holistic criteria

ID	Human label 1	Human label 2	Rel.	Comp.	Dir.	Holistic	
F01	4		3	4	3	3	4

B.2 Barter Deals examples

Example B01

Deal text: We'd love a video capturing the experience of visiting our store in the heart of Amsterdam—from the vibe of the city and the walk over, to exploring the shop itself. While you're there, pick your favorite bag from the collection and make it part of the story!

Keep it authentic, stylish, and feel free to show the charm of the surroundings too!

Applications after 7 days: 39

Product category: Activities

Example B02

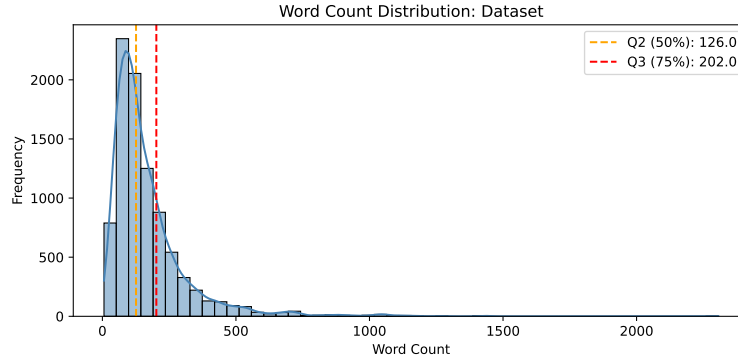
Deal text: Choose 2 coffees and pastries you like:)

Applications after 7 days: 6

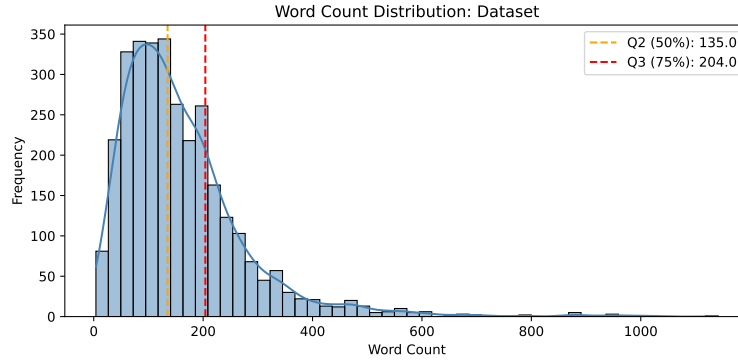
Product category: Food

C Additional Graphs & Diagnostics

C.1 Word Count & Verbosity Bias



(a) FeedbackQA: combined word counts (Question + Answer).



(b) Barter Deals: combined word counts (Title + Body + Requirements).

Fig. 2: Empirical word-count distributions used for the residual verbosity-sensitivity diagnostic. Dashed lines indicate the median and 75th percentile, which define the mature text brackets D_{Q_3} and D_{Q_4} . For FeedbackQA, $Q(0.50) = 126$ and $Q(0.75) = 202$; for Barter Deals, $Q(0.50) = 135$ and $Q(0.75) = 204$.

Table 11: Text-length statistics for Barter Deal text sections

Text Section	Mean Word Count	SD	Median (Q2)	75th Percentile (Q3)
Body	81.89	72.13	64	104
Requirements	74.92	79.55	53	98

C.2 Criterion-Level Rating Distributions

This subsection reports the full criterion-level EV rating distributions used to support the aggregate rating-regime diagnostics in the main text.

Rating Distributions (1-4 Scale): Holistic vs Macro-Formative vs Formative

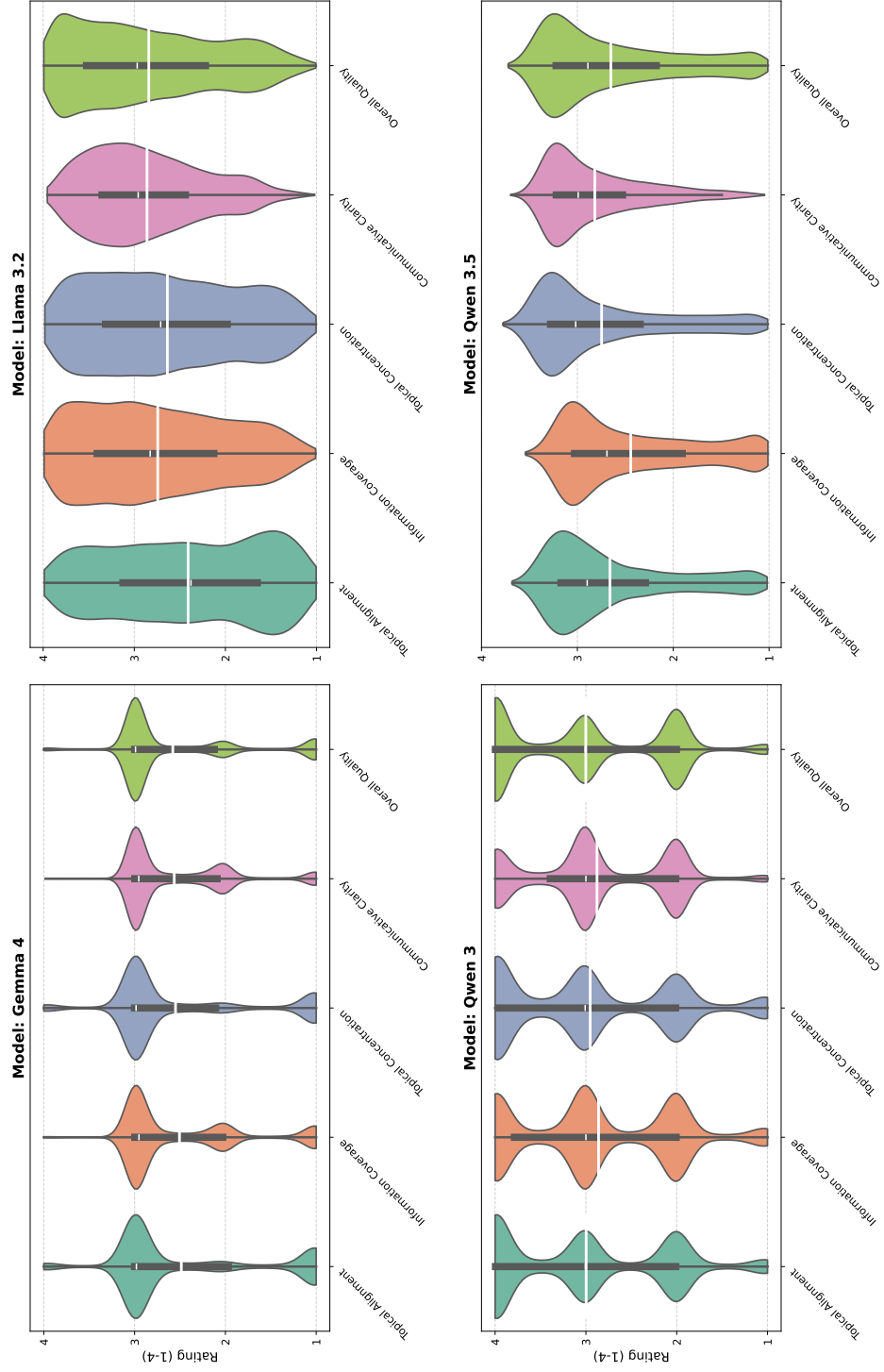


Fig. 3: Criterion-level expected-value rating distributions for Barter Deals.

Rating Distributions (1-7 Scale): Holistic vs Macro-Formative vs Formative

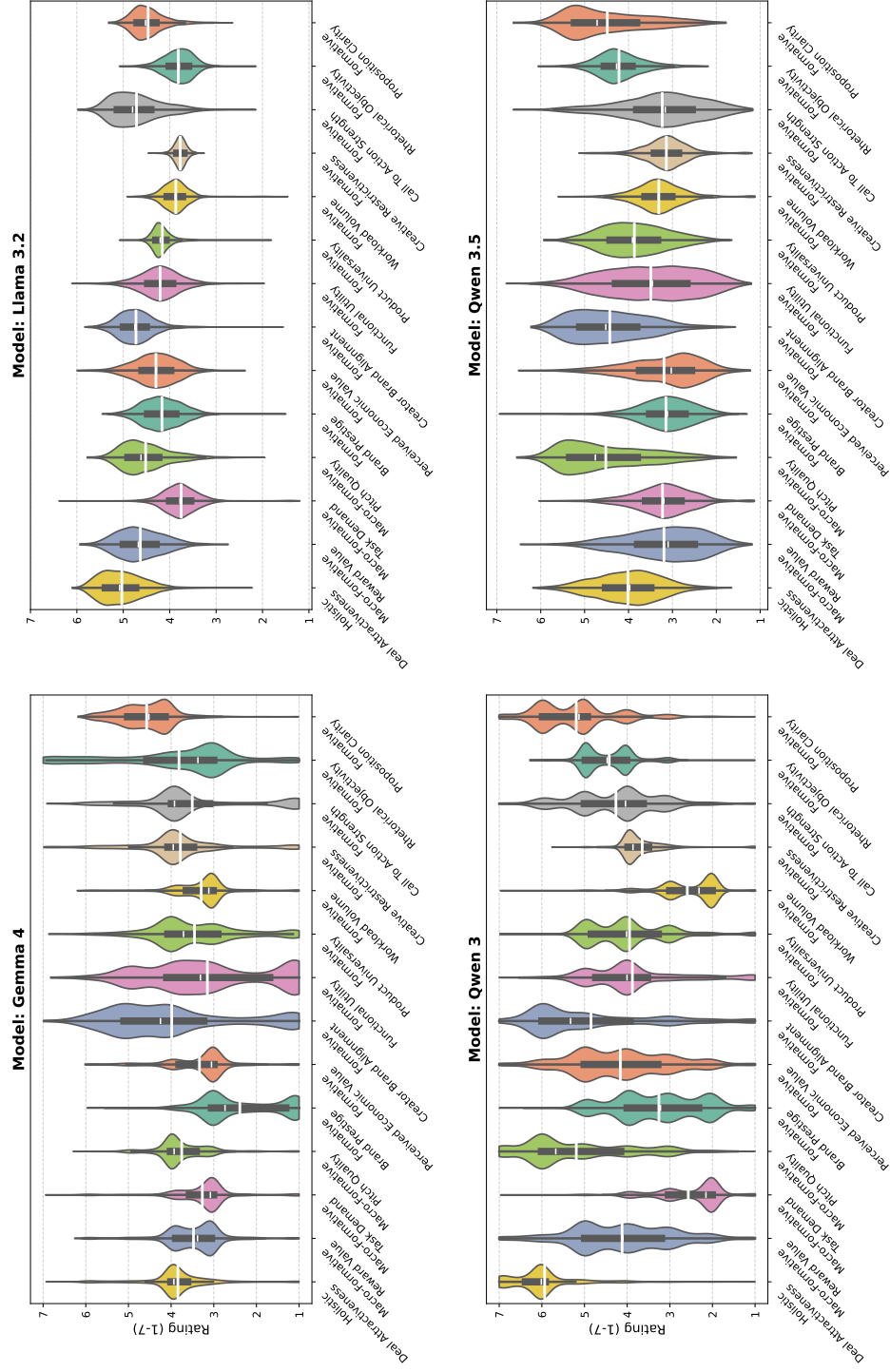


Fig. 4: Criterion-level expected-value rating distributions for Barter Deals.

C.3 Additional Multicollinearity Diagnostics

Table 12: VIF Diagnostics for Barter Deals Formative Specifications

Group	Max VIF	Mean VIF	N VIF > 5	N VIF > 10	Multicollinearity
Gemma 4 Formative	1.79	1.36	0	0	Limited
Gemma 4 Macro-Formative	1.70	1.49	0	0	Limited
Llama 3.2 Formative	3.70	2.54	0	0	Limited
Llama 3.2 Macro-Formative	4.20	2.99	0	0	Limited
Qwen 3 Formative	1.97	1.49	0	0	Limited
Qwen 3 Macro-Formative	1.34	1.25	0	0	Limited
Qwen 3.5 Formative	2.07	1.58	0	0	Limited
Qwen 3.5 Macro-Formative	1.34	1.26	0	0	Limited

Note. VIF values are only computed for the LLM-derived evaluations for the Formative and Macro-Formative conditions, and do not include the non-LLM control variables. Values above 5 are interpreted as moderate multicollinearity and values above 10 as substantial multicollinearity.

Table 13: VIF Diagnostics for Barter Deals Baseline Control Specification

Specification	Max VIF	Mean VIF	$N_{\text{VIF}>5}$	$N_{\text{VIF}>10}$	Multicollinearity
Baseline Controls	5.09	2.04	1	0	Moderate

Note. VIF values are computed for the non-LLM control specification used in the Barter Deals predictive-validity models. The only predictor exceeding 5 is *macro_category_Home Cooking & CPG* with a VIF of 5.09. No predictor exceeds 10.

C.4 Full FeedbackQA Cross-Validation Results

EV-only model results Table 14 reports the full set of cross-validated convergent-validity metrics for the FeedbackQA analysis. The main text reports the compact comparison using RMSE and Spearman’s ρ only; this appendix table additionally reports Pearson’s r and Kendall’s τ . The full results confirm the same substantive pattern: the formative condition consistently outperforms the holistic baselines across model architectures, while the informed holistic condition provides only marginal gains over the naive holistic condition.

Table 14: Full Cross-Validated Convergent Validity Performance on FeedbackQA

Model	Condition	RMSE ↓	Pearson's r ↑	Spearman's ρ ↑	Kendall's τ ↑
Gemma 4	Holistic Naive	1.119	0.440	0.436	0.336
Gemma 4	Holistic Informed	1.111	0.454	0.450	0.347
Gemma 4	Formative	1.094	0.479	0.480	0.370
Llama 3.2	Holistic Naive	1.144	0.396	0.395	0.302
Llama 3.2	Holistic Informed	1.144	0.397	0.396	0.302
Llama 3.2	Formative	1.128	0.424	0.423	0.323
Qwen 3	Holistic Naive	1.044	0.546	0.544	0.421
Qwen 3	Holistic Informed	1.039	0.553	0.550	0.426
Qwen 3	Formative	1.018	0.577	0.578	0.449
Qwen 3.5	Holistic Naive	1.033	0.559	0.559	0.435
Qwen 3.5	Holistic Informed	1.034	0.558	0.558	0.434
Qwen 3.5	Formative	1.019	0.576	0.576	0.449
Human-Human Baseline		1.196	0.542	0.541	0.467

Note. Values report out-of-fold cross-validated performance for EV-only specifications. RMSE measures prediction error; Pearson's r , Spearman's ρ , and Kendall's τ measure alignment with human quality judgments. Lower RMSE and higher correlation coefficients indicate better convergent validity. The human-human baseline compares the two available human annotations and is included as a reference point for annotation consistency, not as a fitted model.

C.5 Full Barter Deals Results

EV-only model results

D Supplementary Measurement Diagnostics

This appendix provides additional methodological detail for the diagnostic procedures used to evaluate the LLM-derived measurements. These diagnostics are not themselves validation tests. Instead, they characterize whether the expected-value (EV) ratings and entropy values behave in ways that are compatible with their intended use as continuous measurement variables.

D.1 Interpreting EV and Entropy Regimes

The distributional diagnostics combine two sources of information: the spread of the EV ratings and the entropy of the restricted rating-token distribution. EV spread indicates how much of the available rating scale is used by the model. Entropy indicates whether the EV score is produced by probability mass concentrated on one or a few rating tokens, or by a smoother distribution over multiple rating tokens.

These metrics allow us to distinguish four rating regimes, which serve as interpretive categories used to compare model behavior.

Broad continuous regime. A broad continuous regime occurs when EV ratings show substantial dispersion across the available rating scale and entropy is not uniformly near zero. In this case, the model differentiates between documents while also producing smooth variation between rating tokens. This is the most desirable regime for a logit-derived continuous measurement variable.

Broad quantized regime. A broad quantized regime occurs when EV ratings vary across the scale, but entropy is consistently low. In this case, the model distinguishes between documents, but it does so primarily through near-argmax selection among discrete rating tokens. The resulting scores behave more like hard ordinal classifications rather than continuous measurements.

Narrow continuous regime. A narrow continuous regime occurs when entropy indicates non-degenerate probability distributions, but EV ratings remain concentrated in a restricted part of the scale. In this case, the model produces smooth variation, but within a compressed interval.

Narrow quantized regime. A narrow quantized regime occurs when EV ratings are compressed and entropy is consistently low. In this case, the model uses only a limited part of the rating scale and assigns probability mass to a small number of rating tokens.

D.2 Residual Verbosity Sensitivity

The residual verbosity diagnostic is designed to distinguish general text-length effects from length effects that remain after documents have reached a mature

length. Not every association between text length and rating is evidence of verbosity bias. Very short texts may receive lower ratings for construct-relevant reasons, because they may lack the information required to satisfy the criterion. The diagnostic therefore compares documents in the upper half of the length distribution, where additional length is less likely to reflect basic informational adequacy.

Let D represent the corpus of evaluated documents and let $Q(p)$ denote the p -th quantile of the word-count distribution. The third and fourth quartile subsets are defined as:

$$D_{Q3} = d \in D \mid Q(0.50) \leq \text{length}(d) \leq Q(0.75),$$

$$D_{Q4} = d \in D \mid Q(0.75) < \text{length}(d) \leq Q(1.00).$$

For a given model m and criterion c_i , let μ_{Q3} and μ_{Q4} denote the mean EV ratings assigned to documents in D_{Q3} and D_{Q4} , respectively. Let s_p denote the pooled standard deviation across the two groups. The signed Q3–Q4 effect is computed as:

$$\delta_{m,c_i} = \frac{\mu_{Q4} - \mu_{Q3}}{s_p}.$$

Positive values indicate that the longest documents receive higher ratings than mature but shorter documents, while negative values indicate that the longest documents receive lower ratings. Because the substantive interpretation of the sign depends on the criterion coding ³, the main diagnostic uses the absolute value $|\delta_{m,c_i}|$. This captures the magnitude of residual length sensitivity regardless of direction.

Following Cohen [9], values around 0.2 are interpreted as small, values around 0.5 as medium, and values of 0.8 or above as large.

D.3 Construct Separability and VIF

For the formative specifications, the LLM-derived predictors are intended to represent distinct components of the target construct. Empirical separability is assessed using the Variance Inflation Factor (VIF). For predictor x_j , the VIF is defined as:

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is obtained by regressing x_j on the remaining predictors. Higher VIF values indicate that a predictor is strongly predictable from the other predictors, which reduces the interpretability of individual coefficients. VIF values

³ For example, Llama 3.2 shows a negative Q3–Q4 effect for *Creative Restrictiveness*. This criterion is inversely coded: higher scores indicate more restrictive requirements, whereas for most other criteria higher scores indicate desirable properties. This illustrates why the absolute effect size is used as the main diagnostic.

below 5 are treated as indicating limited multicollinearity, values between 5 and 10 as moderate multicollinearity, and values above 10 as substantial multicollinearity.

E Evaluation Criteria and Rating Anchors

E.1 FeedbackQA Criteria

Table 15: Formative dimensions of *Q&A Answer Quality*

Dimension	Definition
<i>Topical Alignment</i>	The degree to which the answer addresses the issue and stated constraints of the question.
<i>Information Coverage</i>	The degree to which the answer provides enough relevant information to serve as a full answer.
<i>Topical Concentration</i>	The degree to which the answer remains centered on the asked issue throughout the passage.
<i>Communicative Clarity</i>	The degree to which the answer is expressed in a clear, well-organized, and easy-to-follow way.

E.2 Barter Deals Criteria

Table 16: Fine-grained dimensions of *Deal Attractiveness* mapped to the macro-formative pillars

Dimension	Pillar	Definition
<i>Brand Prestige</i>	Reward	How well-known and trusted the brand name is in the market.
<i>Perceived Economic Value</i>	Reward	The estimated retail value and premium nature of the product, service, or compensation offered to the creator.
<i>Creator Brand Alignment</i>	Reward	How naturally the product, service, or experience fits within a professional content creator’s social media presence.
<i>Functional Utility</i>	Reward	The practical, everyday usefulness and utilitarian value of the physical product or service being offered.
<i>Product Universality</i>	Reward	The mass-market consumer appeal of the physical product or service on offer.
<i>Workload Volume</i>	Cost	The total time and effort required to execute the collaboration.
<i>Creative Restrictiveness</i>	Cost	The level of brand control over the creator’s creative process.
<i>Call-to-Action Strength</i>	Pitch Quality	The level of excitement and warmth in the brand’s invitation to collaborate.
<i>Rhetorical Objectivity</i>	Pitch Quality	The level of factual realism versus promotional marketing hype in the pitch.
<i>Proposition Clarity</i>	Pitch Quality	The structural clarity and cognitive ease of processing the pitch.